Electrothermal Manipulation of Abrikosov Vortices via Localized Heating

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Motivation

- Topological superconductors may host Majorana zero modes (MZMs) inside Abrikosov vortex cores.
- Future quantum computation requires controlled vortex manipulation for possible braiding operations.
- However, directly manipulating and observing Majorana modes remains experimentally challenging.

Question:

Can we control vortex motion using localized electrothermal effects?

Research Idea

Bottom electrode $\rightarrow$ Joule heating $\rightarrow$ Temperature gradient $\rightarrow$ Vortex motion

The project focuses on three questions:

1. Can a localized hotspot capture a vortex?
2. Can two hotspots transfer a vortex?
3. Can programmable hotspots guide vortex motion?

Physical Model

The local temperature distribution is approximated by Gaussian hotspots:

$$T(\mathbf{r}, t) = T_0 + \sum_k \Delta T_k(t) \exp\left(-\frac{|\mathbf{r} - \mathbf{R}_k|^2}{2\sigma_T^2}\right)$$

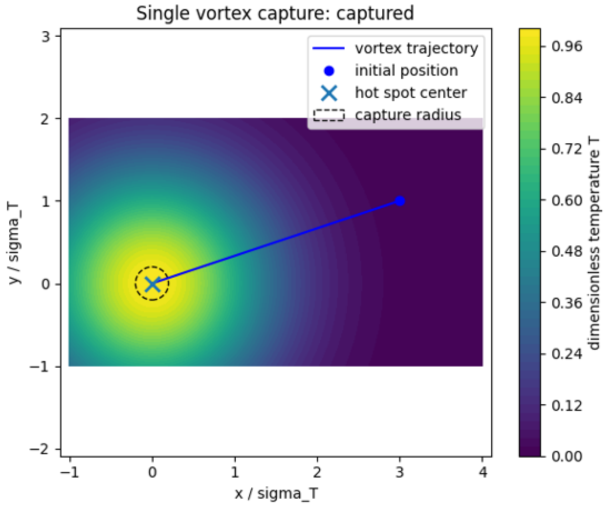
The vortex dynamics follows an overdamped equation:

$$\eta \frac{d\mathbf{r}}{dt} = \mathbf{f}_T + \mathbf{f}_{pin} + \mathbf{f}_{vv}$$

where:

- $\mathbf{f}_T$: thermal driving force
- $\mathbf{f}_{pin}$: pinning force
- $\mathbf{f}_{vv}$: vortex-vortex interaction

Single Hotspot Capture

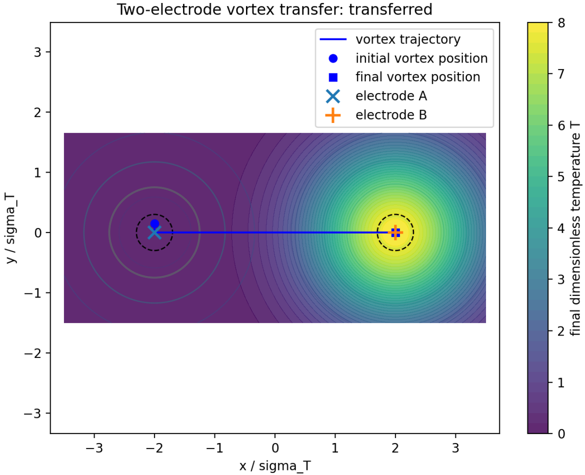


A vortex initially far from the hotspot is attracted toward the high temperature region.

Result:

- Single hotspot can capture vortex.
- Stronger heating gives larger capture radius.

Vortex Transfer Between Two Electrodes



Heating protocol:

$$\Delta T_A(t) \downarrow$$

$$\Delta T_B(t) \uparrow$$

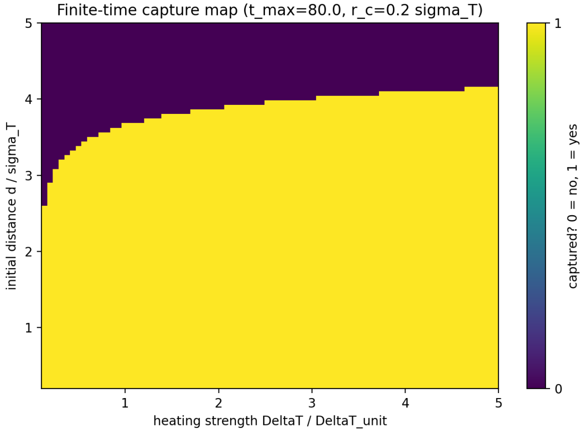
The vortex follows the moving temperature landscape.

Result:
Controlled vortex transfer is achievable.

Parameter Dependence

Successful transfer depends on:

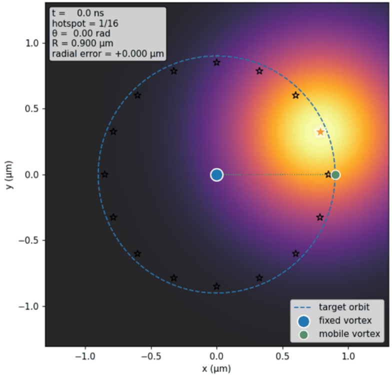
- Heating strength ΔT
- Distance between electrodes
- Switching time



Programmable Hotspot Array

To guide vortex motion:

- Multiple hotspots are arranged along a circular path.
- Hotspots are activated sequentially.
- A moving vortex follows the effective potential landscape.



Physical Time Scale Estimation

Using $NbSe_2$ parameters:

$$\tau = \frac{\eta\sigma^2}{\alpha_T \Delta T}$$

gives:

$$\tau \approx 7\ ns$$

Therefore:

$$t_{\mathrm{on}} \approx 55\ ns$$

is sufficient for vortex following.

The estimated timescale is physically reasonable, although realistic devices require more detailed modeling.

Conclusion and Future Work

Conclusion

- Developed a 2D overdamped vortex dynamics model.
- Demonstrated:
  - vortex capture
  - vortex transfer
  - circular vortex motion
- Electrothermal manipulation provides a possible route toward programmable vortex control.

Future work

- Solve realistic heat diffusion equation.
- Include stronger pinning effects.
- Compare with experimental parameters.

Appendix: Thermal Model

The temperature field generated by localized heating is approximated as Gaussian:

$$T(\mathbf{r}, t) = T_0 + \sum_k \Delta T_k(t) \exp\left(-\frac{|\mathbf{r} - \mathbf{R}_k|^2}{2\sigma_T^2}\right)$$

where:

- T_0: background temperature
- ΔT_k: temperature increase of hotspot k
- $\mathbf{R}_k$: hotspot position
- σ_T: hotspot width

Appendix: Thermal Driving Force

The vortex prefers regions with reduced condensation energy.
The effective thermal potential is approximated as:

$$U_T(\mathbf{r}) = -\alpha_T(T(\mathbf{r}) - T_0)$$

Therefore,

$$\mathbf{f}_T = -\nabla U_T = \alpha_T \nabla T$$

For a Gaussian hotspot:

$$\nabla T = -\frac{\Delta T}{\sigma_T^2}(\mathbf{r} - \mathbf{R}_h) \exp\left(-\frac{|\mathbf{r} - \mathbf{R}_h|^2}{2\sigma_T^2}\right)$$

Appendix: Vortex Dynamics

The vortex motion is assumed to be overdamped:

$$\eta \frac{d\mathbf{r}}{dt} = \mathbf{f}_T + \mathbf{f}_{pin} + \mathbf{f}_{vv}$$

The inertial term is neglected because vortex motion occurs in the strongly dissipative regime.
The Bardeen-Stephen model gives:

$$\eta = \frac{\Phi_0 B_{c2}}{\rho_n}$$

Appendix: Response Time

Near the hotspot center:

$$|\mathbf{r} - \mathbf{R}_h| \ll \sigma_T$$

The thermal force becomes approximately linear:

$$\mathbf{f}_T \approx -k_l(\mathbf{r} - \mathbf{R}_h)$$

where:

$$k_l = \alpha_T \frac{\Delta T}{\sigma_T^2}$$

Appendix: Response Time (Continued)

The equation becomes:

$$\eta \frac{d\mathbf{u}}{dt} = -k_l \mathbf{u}$$

with:

$$\mathbf{u}(t) = \mathbf{u}_0 e^{-t/\tau}$$

and:

$$\tau = \frac{\eta}{k_l} = \frac{\eta \sigma_T^2}{\alpha_T \Delta T}$$

Appendix: Estimation of Switching Time

The vortex distance follows:

$$r(t) = de^{-t/\tau}$$

To enter the capture radius:

$$r(t_{on}) \leq r_c$$

Therefore:

$$t_{on} \geq \tau \ln \frac{d}{r_c}$$

Appendix: Estimation of Switching Time (Continued)

Using:

$$r_c = 0.10\mu m, \quad d = 0.80\mu m$$

gives:

$$t_{on} \approx 11.4ns$$

A conservative value:

$$t_{on} = 55ns$$

is used in simulation.

Appendix: Vortex-Vortex Interaction

For multiple vortices, London theory gives:

$$U_{vv}(R) \propto K_0\left(\frac{R}{\lambda}\right)$$

where K_0 is the modified Bessel function.
The interaction force is:

$$\mathbf{f}_{vv} = -\nabla U_{vv}$$

This produces vortex-vortex repulsion.

Appendix: Choice of Orbit Parameters

Constraints:

- Avoid vortex core overlap:

$$R_{orbit} \gg \xi_0$$

- Avoid excessive vortex repulsion.
- Maintain sufficient hotspot attraction.

Chosen parameters:

$$R_{orbit} = 0.90\mu m$$

$$N = 16$$

$$d_{hot} = 2R_{orbit} \sin \frac{\pi}{N} = 0.345\mu m$$

Appendix: Material Parameters

Parameters estimated from $NbSe_2$:

$$T_c = 7.2K$$

$$\xi_0 = 10nm$$

$$\lambda_0 = 200nm$$

$$B_{c2} = 3T$$

$$\rho_n = 1.0 \times 10^{-7}\Omega m$$

These values are used only for order-of-magnitude estimation.

References

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